SPECIFICATION SHEET | PAIC

Portable Autonomous Imaging Cytometer

Automatizes the identification and monitoring of aquatic microparticles

PRODUCT OVERVIEW

Identifying aquatic microparticles should not be such a daunting task. The PAIC offers a scalable solution to identify aquatic microparticles and follow their concentration over multiple weeks. It continually detects and identifies the microscopic species found in water and warns the user when a dangerous situation is found. It is easy to deploy and operates for up to two weeks at a time without human intervention.



The Portable Autonomous Imaging Cytometer (PAIC)

Applications

- **Early intervention of algal blooms:** be alerted of a potential algal bloom as soon as the concentration of a specific algal specie exceeds a safe threshold.
- **Microplastics regulation:** monitor and quantify different types of microplastics in industrial effluents.
- **Track invasive species:** enables early detection of invasive species in protected lakes and rivers across the seasons.
- **Environmental research:** characterizes temporal changes in aquatic microbial biodiversity and quantifies the effects of environmental parameters.

WHY CUSTOMERS CHOOSE IT

- **Fully autonomous:** can operate for 2 weeks at a time without any human intervention.
- **Scalable:** its low price-point enables distributed analysis across multiple locations on the same lake, allowing the quantification of geographical and environmental factors.
- **Portable:** can be brought easily on-site!
- **Plug-and-play:** comes pre-calibrated from the manufacturer, no further calibration required; easy to use and repair.



PAIC deployed on a research buoy [projected June 2027!]

TECHNICAL SPECIFICATIONS			
Dimensions	25 cm long x 12 cm wide	**Weight**	1 kg
Dimensions with metal case	35 cm long × 14 cm wide	**Weight with metal case**	10 kg
Power connectivity	USB-C 36W – PoE+ 30W	**Data connectivity**	USB 3.0 – Ethernet – RS-232
Interface	PC – Android – IOS	**Battery life**	2 hours
Power needs (active/sleep)	(12 W / 500 mW)	**Image size**	1200 x 1920 pixels @ 60 fps
Size detection range	10 μm – 500 μm	**Pixel size**	RGB CMOS 3 μm x 3 μm
Operating temperatures	-20 °C – 85 °C	**Water flow rate**	200 mL/h

Ready to see it in action?

Talk with one of our representatives.

karim.bouzid.1@ulaval.ca
+1 819-760-0099

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